How to Use This Guide

Use this guide to quickly set up your Enterasys® B5 switch. Enterasys Networks recommends that you refer to the *Enterasys B5 Gigabit Ethernet Switch Hardware Installation Guide* for specifications and safety warnings.

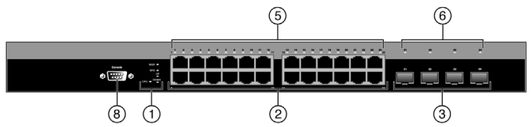
The latest B5 documentation is located at www.enterasys.com/support/manuals.

Hardware Components

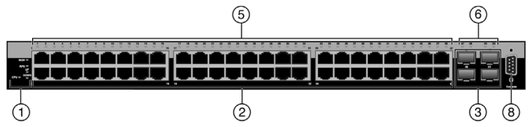
The following figures show the several types of front panel port and rear panel connections on your B5 switch.

Figure 1 B5 Switch Front Panels

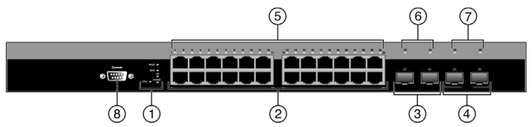
B5G124-24, B5G124-24P2



B5G124-48, B5G124-48P2



B5K125-24, B5K125-24P2



B5K125-48, B5K125-48P2

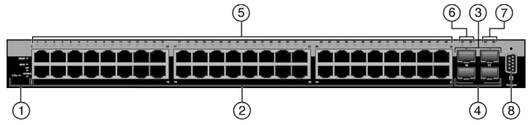
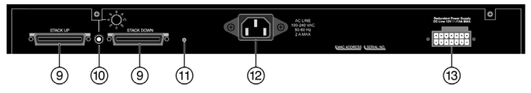
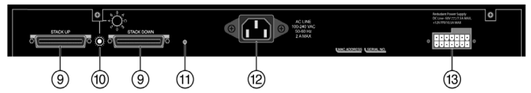


Figure 2 B5 Switch Back Panels

Non-PoE B5 Switches



PoE B5 Switches



Key

1	Switch status LEDs	8	Console port
2	RJ45 10/100/1000 Mbps ports	9	Stack connection ports
3	SFP slots	10	Rotary switch
4	10G slots	11	Password reset button
5	RJ45 status LEDs	12	AC power input connector
6	SFP port status LEDs	13	Redundant power supply (RPS) connector
7	10G port status LEDs		

Handling the Switch

Caution: The switch can be damaged by electrostatic discharge.

To prevent electrostatic damage, attach an electrostatic discharge (ESD) wrist strap to your wrist before handling the switch.

Unpack the switch as follows:

1. Remove the packing material protecting the switch.
2. Remove the tape seal on the non-conductive bag to remove the switch.
3. Perform a visual inspection of the switch for any signs of physical damage. Contact Enterasys Networks if there are any signs of damage. See "Getting Help" for more information on contacting Enterasys Networks.

Stacking Switches

Up to eight B5 switches can be stacked together and connected by high-speed stacking cables, which allow the entire stack to operate with a single IP address.

The B5 supports the following high-speed stacking cables only, which you must order separately:

- STK-CAB-SHORT
- STK-CAB-LONG
- STK-CAB-2M
- STK-CAB-5M

Note: You can stack a B5 switch only with other B5 switches.

For complete information about stacking, refer to the *Enterasys B5 Gigabit Ethernet Switch Hardware Installation Guide.*

Connecting Power to the Switch

Electrical Hazard: Only qualified personnel should perform installation procedures.

You can connect to a single, primary source of power, or to two sources of power for redundancy. The example used here describes connecting to two power sources.

The following redundant power supplies are available to be purchased from Enterasys Networks for connection to a B5 switch:

- STK-RPS-150PS, a 102-156 watt DC power supply capable of providing power to a fully-loaded non-PoE switch.
- STK-RPS-500PS, a 500 watt DC power supply recommended for full PoE capability.

To power-up your B5 switch:

1. Attach the power cord from your redundant power supply into the B5 switch's redundant power receptacle.
2. Attach the AC power cord to the B5 switch's AC power receptacle.
3. Plug the redundant power supply and the B5 switch AC power cords into dedicated, grounded AC outlets.
4. Once power is connected, verify that the CPU (system) LED turns amber until the B5 switch finishes its initialization.

If the initialization process is successful, the CPU LED turns green. If the CPU LED does not turn green, refer to the *Enterasys B5 Gigabit Ethernet Switch Hardware Installation Guide* for troubleshooting information.

Completing Switch Setup and Downloading the Latest Firmware

Once you have connected power to the B5 switch and verified LED activity, you can complete the setup process as follows. Refer to the *Enterasys B5 Gigabit Ethernet Switch Hardware Installation Guide* for more information.

1. Determine the latest B5 firmware version by visiting the Enterasys Networks download site at: http://secure.enterasys.com/services/support/downloads/software
2. Connect the switch to the network.
3. Connect a management station to the console port.
4. Verify that the network devices connected to the switch ports are powered on, and that each link/activity LED is on (solid green or blinking green).
5. At the device connected to the console port, perform the following:
   a. Enter **admin** for Username.
   b. At the Password prompt, press **ENTER** (RETURN).
   c. At the command prompt, determine if the latest firmware image is loaded on the switch by entering this command:
      `show version`
   d. If the output (under **FW Version**) displays an older version number than that determined in step 1, download and activate the new version as directed on the download website. (Alternatively, you can use the CLI commands listed in Table 1.)

Basic Setup Commands

Table 1 lists CLI commands that are required for setting up the B5 switch with the latest firmware. Table 2 lists additional configuration commands for your B5 switch. For the complete list of CLI commands, see the *Enterasys B5 CLI Reference*.

Table 1 Required CLI Setup Commands

Step	Task	CLI commands
1	Set a new password	**set password** [*username*]
2	Set the switch IP address	**set ip address** *ip-address* [**mask** *ip-mask*] [**gateway** *ip-gateway*]
3	Download, activate, and verify new firmware on the switch using TFTP copy **Note:** You can reboot the system immediately with the **set boot system** command, or reboot later with the **reset** command.	**copy tftp://***tftp_server_ip_address*/*filename* **system:image** **set boot system** *filename* **show version**

Table 2 Optional CLI Setup Commands

Task	CLI commands
Save the active configuration	**save config**
Enable or disable SSH	**set ssh enable | disable**
Enable or disable Telnet	**set telnet {enable | disable} [inbound | outbound | all]**
Enable or disable HTTP management (WebView)	**set webview {enable | disable}**
Enable or disable SNMP port link traps	**set port trap** *port-string* **{enable | disable}**
Set the per port broadcast limit	**set port broadcast** *port-string threshold-value*
Configure a VLAN	**set vlan create** *vlan-id* **set port vlan** *port-string vlan-id* **modify-egress**
Set a Syslog server IP and severity	**set logging server** *index* **ip-addr** *ip-addr* **severity** *severity* **state enable**
Configure and enable a RADIUS server	**set radius server** *index ip-addr port* [*secret-value*] **{realm {management-access | any | network-access}** **set radius enable**

Specifications

For a complete list of specifications, see the *Enterasys B5 Gigabit Ethernet Switch Hardware Installation Guide*.

Interfaces

Table 3 B5 Interface Descriptions

B5G124-24	• 24 10/100/1000Base-T ports • Four combo SFP ports
B5G124-24P2	• 24 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Four combo SFP ports
B5G124-48	• 48 10/100/1000Base-T ports • Four combo SFP ports
B5G124-48P2	• 48 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Four combo SFP ports
B5K125-24	• 24 10/100/1000Base-T ports • Two combo SFP ports • Two 10G ports
B5K125-24P2	• 24 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Two combo SFP ports • Two 10G ports
B5K125-48	• 48 10/100/1000Base-T ports • Two combo SFP ports • Two 10G ports
B5K125-48P2	• 48 10/100/1000Base-T 802.3af and 802.3at PoE-capable ports • Two combo SFP ports • Two 10G ports

Switch Dimensions

Size: 44.1 W x 4.4 H x 36.85 D cm
17.4 W x 1.7 H x 14.5 D in

Power Consumption

Input Voltage: 100 to 240VAC

Temperature and Humidity

Operating temperature: 0°C to 50°C

Storage temperature: -40°C to +70°C

Operating relative humidity: 5% to 95%

Getting Help

World Wide Web	www.enterasys.com/support/
Phone	1-800-872-8440 (toll-free in U.S. and Canada) or 1-978-684-1888 To find the Enterasys Networks Support toll-free number in your country: www.enterasys.com/services/support/contact/
Internet mail	support@enterasys.com To expedite your message, type **[Switching]** in the subject field of your message.
Latest image and release notes	www.enterasys.com/services/support/downloads/software/

Related Documents

The latest B5 documentation is located online at:
www.enterasys.com/support/manuals

Warranty

Warranty information for the B5 switch is located online at:

www.enterasys.com/support/warranty.aspx

www.enterasys.com/company/literature/enterasys-lw-ds.pdf

Notice

Enterasys Networks reserves the right to make changes in specifications and other information contained in this document and its web site without prior notice. The reader should in all cases consult Enterasys Networks to determine whether any such changes have been made.

The hardware, firmware, or software described in this document is subject to change without notice.

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Andover, MA 01810

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Part Number: 9034512-01 February 2010

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All other product names mentioned in this manual may be trademarks or registered trademarks of their respective companies.

Regulatory Compliance Information

Federal Communications Commission (FCC) Notice

This device complies with Part 15 of the FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

NOTE: This equipment has been tested and found to comply with the limits for a class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment uses, generates, and can radiate radio frequency energy and if not installed in accordance with the operator's manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause interference in which case the user will be required to correct the interference at his own expense.

WARNING: Changes or modifications made to this device which are not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Industry Canada Notice

This digital apparatus does not exceed the class A limits for radio noise emissions from digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications.

Le présent appareil numérique n'émet pas de bruits radioélectriques dépassant les limites applicables aux appareils numériques de la class A prescrites dans le Règlement sur le brouillage radioélectrique édicté par le ministère des Communications du Canada.

Class A ITE Notice

WARNING: This is a Class A product. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

Clase A. Aviso de ITE

ADVERTENCIA: Este es un producto de Clase A. En un ambiente doméstico este producto puede causar interferencia de radio en cuyo caso puede ser requerido tomar medidas adecuadas.

Klasse A ITE Anmerkung

WARNHINWEIS: Dieses Produkt zählt zur Klasse A (Industriebereich). In Wohnbereichen kann es hierdurch zu Funkstörungen kommen, daher sollten angemessene Vorkehrungen zum Schutz getroffen werden.

VCCI Notice

This is a class A product based on the standard of the Voluntary Control Council for Interference by Information Technology Equipment (VCCI). If this equipment is used in a domestic environment, radio disturbance may arise. When such trouble occurs, the user may be required to take corrective actions.

この装置は，情報処理装置等電波障害自主規制協議会（ＶＣＣＩ）の基準に基づくクラスＡ情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。

BSMI EMC Statement — Taiwan

This is a class A product. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

警告使用者：
這是甲類的資訊產品，在居住的環境中使用時，可能會造成射頻干擾，在這種請況下，使用者會被要求採取某些適當的對策。

AS/NZS CISPR 22

Hazardous Substances

This product complies with the requirements of European Directive, 2002/95/EC, Restriction of Hazardous Substances (RoHS) in Electrical and Electronic Equipment.

European Waste Electrical and Electronic Equipment (WEEE) Notice



In accordance with Directive 2002/96/EC of the European Parliament on waste electrical and electronic equipment (WEEE):

1. The symbol above indicates that separate collection of electrical and electronic equipment is required and that this product was placed on the European market after August 13, 2005, the date of enforcement for Directive 2002/96/EC.
2. When this product has reached the end of its serviceable life, it cannot be disposed of as unsorted municipal waste. It must be collected and treated separately.
3. It has been determined by the European Parliament that there are potential negative effects on the environment and human health as a result of the presence of hazardous substances in electrical and electronic equipment.
4. It is the users' responsibility to utilize the available collection system to ensure WEEE is properly treated.
   For information about the available collection system, please go to www.enterasys.com/services/support/ or contact Enterasys Customer Support at 353 61 705586 (Ireland).

产品说明书附件
Supplement to Product Instructions

部件名称 (Parts)	有毒有害物质或元素 (Hazardous Substance)					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr^{6+})	多溴联苯 (PBB)	多溴二苯醚 (PBDE)
金属部件 (Metal Parts)	×	○	○	○	○	○
电路模块 (Circuit Modules)	×	○	○	○	○	○
电缆及电缆组件 (Cables & Cable Assemblies)	×	○	○	○	○	○
塑料和聚合物部件 (Plastic and Polymeric parts)	○	○	○	○	○	○
电路开关 (Circuit Breakers)	○	○	○	○	○	○

○： 表示该有毒有害物质在该部件所有均质材料中的含量均在 SJ/T 11363-2006 标准规定的限量要求以下。
Indicates that the concentration of the hazardous substance in all homogeneous materials in the parts is below the relevant threshold of the SJ/T 11363-2006 standard.

×： 表示该有毒有害物质至少在该部件的某一均质材料中的含量超出SJ/T 11363-2006 标准规定的限量要求。
Indicates that the concentration of the hazardous substance of at least one of all homogeneous materials in the parts is above the relevant threshold of the SJ/T 11363-2006 standard.

对销售之日的所售产品，本表显示，
凯创供应链的电子信息产品可能包含这些物质。注意：在所售产品中可能会也可能不会含有所有所列的部件。
This table shows where these substances may be found in the supply chain of Enterasys' electronic information products, as of the date of sale of the enclosed product. Note that some of the component types listed above may or may not be a part of the enclosed product.

除非另外特别的标注，此标志为针对所涉及产品的环保使用期标志．某些零部件会有一个不同的环保使用期(例如，电池单元模块)贴在其产品上．
此环保使用期限只适用于产品是在产品手册中所规定的条件下工作．
The Environmentally Friendly Use Period (EFUP) for all enclosed products and their parts are per the symbol shown here, unless otherwise marked. Certain parts may have a different EFUP (for example, battery modules) and so are marked to reflect such. The Environmentally Friendly Use Period is valid only when the product is operated under the conditions defined in the product manual.

Safety Information
Class 1 Laser Transceivers

The single mode interface modules use Class 1 laser transceivers. Read the following safety information before installing or operating these modules.

The Class 1 laser transceivers use an optical feedback loop to maintain Class 1 operation limits. This control loop eliminates the need for maintenance checks or adjustments. The output is factory set, and does not allow any user adjustment. Class 1 Laser transceivers comply with the following safety standards:

- 21 CFR 1040.10 and 1040.11 U.S. Department of Health and Human Services (FDA).
- IEC Publication 825 (International Electrotechnical Commission).
- CENELEC EN 60825 (European Committee for Electrotechnical Standardization).

When operating within their performance limitations, laser transceiver output meets the Class 1 accessible emission limit of all three standards. Class 1 levels of laser radiation are not considered hazardous.

When the connector is in place, all laser radiation remains within the fiber. The maximum amount of radiant power exiting the fiber (under normal conditions) is -12.6 dBm or 55 x 10^{-6} watts.

Removing the optical connector from the transceiver allows laser radiation to emit directly from the optical port. The maximum radiance from the optical port (under worst case conditions) is 0.8 W cm^{-2} or 8 x 10^{3} W m^{2} sr-1.

Do not use optical instruments to view the laser output. The use of optical instruments to view laser output increases eye hazard. When viewing the output optical port, power must be removed from the network adapter.

Safety Compliance

Warning: Fiber Optic Port Safety

CLASS I LASER DEVICE

When using a fiber optic media expansion module, never look at the transmit laser while it is powered on. Also, never look directly at the fiber TX port and fiber cable ends when they are powered on.

Avertissment: Ports pour fibres optiques - sécurité sur le plan optique

DISPOSITIF LASER DE CLASSE I

Ne regardez jamais le laser tant qu'il est sous tension. Ne regardez jamais directement le port TX (Transmission) à fibres optiques et les embouts de câbles à fibres optiques tant qu'ils sont sous tension.

Warnhinweis: Faseroptikanschlüsse - Optische Sicherheit

LASERGERÄT DER KLASSE I

Niemals ein Übertragungslaser betrachten, während dieses eingeschaltet ist. Niemals direkt auf den Faser-TX-Anschluß und auf die Faserkabelenden schauen, während diese eingeschaltet sind.

Declaration of Conformity

Application of Council Directive(s): **2004/108/EC**
2006/95/EC

Manufacturer's Name: Enterasys Networks, Inc.

Manufacturer's Address: **50 Minuteman Road**
Andover, MA 01810
USA

European Representative Address: Enterasys Networks **Ltd.**
Nexus House, Newbury
Business Park
London Road, Newbury
Berkshire RG14 2PZ, England

Conformance to Directive(s)/Product Standards: **EC Directive 2004/108/EC**
EN 55022:2006
EN 55024:1998
EN 61000-3-2:2006
EN 61000-3-3:1995
EC Directive 2006/95/EC
EN 60950-1:2006
EN 60825-1:2007
EN 60825-2:2004

Equipment Type/Environment: **Information Technology Equipment, for use in a Commercial or Light Industrial Environment.**

Enterasys Networks, Inc. declares that the equipment packaged with this notice conforms to the above directives.

Enterasys® B5
Gigabit Ethernet Switch

Quick Reference

B5G124-24, B5G124-24P2
B5G124-48, B5G124-48P2
B5K125-24, B5K125-24P2
B5K125-48, B5K125-48P2

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